

CM 7028 Digital Media Production



Chutisant Kerdvibulvech, Ph.D.
Assistant Professor

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นน 7028 การผลิตสื่อดิจิทัล

L1

INSTRUCTOR

Asst. Prof. Dr. Chutisant Kerdvibulvech

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 - Office: 18th Floor of Navamindradhiraj Building
 - Office Hour: Mon 13.00-16.00
- (Please Email/Facebook me first to make sure that I will be there.)

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Course Syllabus and Lectures

All course syllabus and lectures can be downloaded online via the following links.

- Course Syllabus
 - <http://gscm.nida.ac.th/~chutisant/CM7028/CS.pdf>
- Lectures
 - <http://gscm.nida.ac.th/~chutisant/CM7028/L1.pdf>
 - <http://gscm.nida.ac.th/~chutisant/CM7028/L2.pdf>
 - <http://gscm.nida.ac.th/~chutisant/CM7028/L3.pdf>
 - etc

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Evaluation

- Assignments/Quiz, Participation and Attendance
20 %
- Midterm Project
40 %
- Final Project
40 %

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Textbooks

- Mixed Reality: Merging Real and Virtual Worlds, Yuichi Ohta, Hideyuki Tamura, 418 pages, Springer, ISBN-13: 978-3642875144 (April 15, 2014)
- Augmented Reality: A Modern Vision-Based Approach, Author: Chutisant Kerdvibulvech, Publish by: Rangsit University Publishing, Printed in: Thailand by S Charoen Printing Co., Ltd, ISBN 978-616-7687-21-6, 1st Edition (2014)
- Computer Vision: Models, Learning, and Inference, Simon J. D. Prince, Cambridge University Press; 1 edition (June 18, 2012)
- Augmented Reality: A Practical Guide, Stephen Cawood, Mark Fiala, ISBN-10: 1934356034, (2008)

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Academic Papers (Scopus)

- C. Kerdvibulvech, "Vision and Virtual-based Human Computer Interaction Applications for a New Digital Media Visualization", 23rd International Conference on Computer Graphics, Visualization and Computer Vision 2015 (WSCG), Eurographics Association & ACM SIGGRAPH (Scopus, SCImago and Thomson Reuters ISI), Plzen, Czech Republic, June 8 - 12, 2015.
- C. Kerdvibulvech, "An Innovative Use of Multidisciplinary Applications Between Information Technology and Socially Digital Media for Connecting People", The Multidisciplinary International Social Networks Conference 2015 (Springer), in Communication of Computer and Information Science (CCIS), (Scopus and SCImago Quartile 4th), Ehime, Japan, September 1 – 3, 2015.
- C. Kerdvibulvech, "A Methodology for Hand and Fingers Motion Analysis Using Adaptive Probabilistic Models," EURASIP Journal on Embedded Systems (JES), 9 pages, No. 18, Springer Publisher (Elsevier's Scopus, EiCompendex, INSPEC, SciFinder, SCImago, ACM, DBLP, and DOAJ), 2014.
- C. Kerdvibulvech and K. Yamauchi, "Structural Human Shape Analysis for Modeling and Recognition," Joint IAPR International Workshops on Structural and Syntactic Pattern Recognition (S+SSPR'14), Lecture Notes in Computer Science (LNCS), vol. 8621, Joensuu, Finland, Springer-Verlag Berlin Heidelberg, pp. 282–290, August 20-22, 2014.

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Websites and Links

- <http://ieeexplore.ieee.org/>
- <http://dl.acm.org/>
- <http://link.springer.com/bookseries/558>
- <http://www.dailynews.co.th/>
- <http://www.ecommerce-magazine.com/>
- <http://www.manager.co.th>

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Overview

- Background
- Digital Media by New Technologies
- 3D Display / 3D Television
- Augmented Reality

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Education



www.chula.ac.th/

- B.E. (Hons) in Computer Engineering
Chulalongkorn University



www.keio.ac.jp

- MSc. / Ph.D. in Engineering, Department of Information and
Computer Science
Keio University, the oldest university in Japan

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Academic Work Experience

- Teaching and being as a research committee in following
universities, such as
 - Rangsit University, Sripatum University, PIM
 - Chulalongkorn University, Mahidol University, Kasetsart
University
 - NIDA
- Technical Committees for many academic international
conferences and journals (Scopus and Scimago)

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Media Work Experience (1/3)

1. Newspapers

- IT Columnist for Dailynews newspaper, every Wednesday (page 23)



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Media Work Experience (2/3)

2. Magazines

- Columnist for Manager 360° Magazine, iTech 360°
- Columnist for e-commerce Magazine, Augmented Reality



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Media Work Experience (3/3)

3. Televisions

- IT News Analyst for Nation Channel TV Program



- IT News Analyst for RSU TV Channel



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Digital Media by New Technologies

Digital Media

Digital media are any media that are encoded in a machine-readable format. [*Technology Brief*. University of Guelph.]

1. Digital Imagery / Digital Video
2. Web Pages / Websites
3. Digital Audio, e.g. mp3s
4. e-books
5. Social Media



www.linkedin.com

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Important Role of Digital Media

Online video has become ubiquitous, digital media consumption has become the norm.

Facts about Digital Media

1. Viewers are heavily engaged with digital media – less so with linear television and newspaper.
2. Television and newspaper still matter, but it can no longer succeed as a stand alone platform.



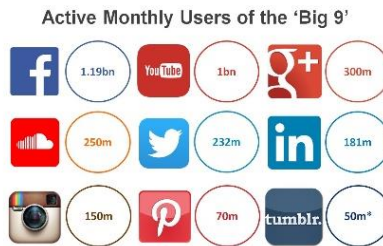
::Source::
Digital Media Trends and the Future of
Television, Dr. Seth Geiger,
President SmithGeiger LLC (CA, USA)

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<https://www.linkedin.com/>

Trends for Digital Media Trends for 2014

Active monthly users of the 'big 9'



As at 15 November 2013 (multiple sources cited)
* Tumblr does not release active monthly user data and this figure is based on an estimate from
 Buzzmonitor.com. It does appear that the current 150m daily active users figure is based on daily active users.

kamber

::Source::
 Kamber (Sydney, Australia)

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kamber

14 Social Media Trends for 2014

Kamber.com.au

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Mobile Phone As Home Computer

- Mobile Phone as PC
- Mobile phones are the new computer
- Designing website/social media for mobile phones is undoubtedly essential



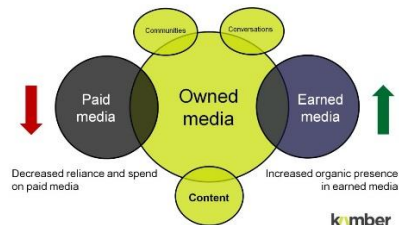
voices.suntimes.com

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Paid Media / Earned Media

The content marketing argument



kumber

::Source::
Kamber (Sydney, Australia)

kumber

14 Social Media Trends for 2014

Kamber.com.au

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Creative-Digital Economy

- Placement: Its where young (and old) people are
- Price to invest: Minimal
- But..
 - Easy to build innovatively
 - Create HUGE profits



<http://www.dailynews.co.th/>

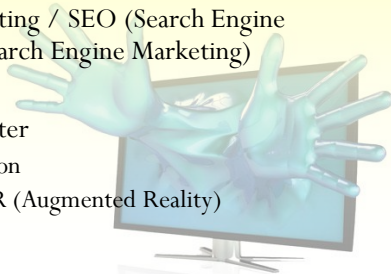
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The Future of Digital Media

Corporate Image and Reputation Management using Technologies of the Future

- Big Data / Cloud Computing / SEO (Search Engine Optimization) – SEM (Search Engine Marketing)
- Mobile-Oriented
- Reaching-Out-of-Computer
 - 3D-Display / 3D Television
 - MR (Mixed Reality) / AR (Augmented Reality)



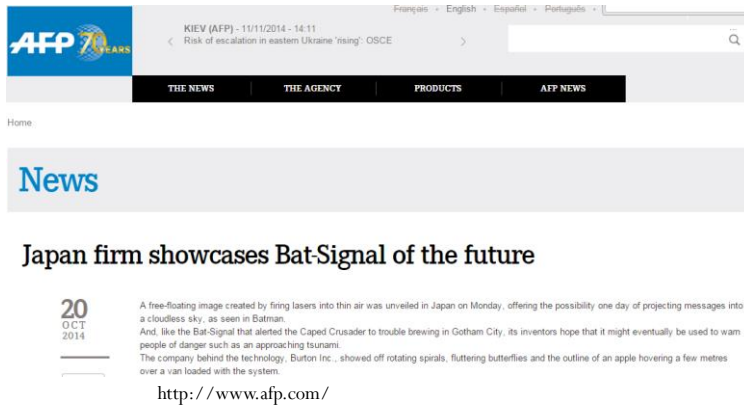
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www.3dsg.com.au

3D Display / 3D Television

3D Display / 3D Television



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3D Display / 3D Television

A free-floating image created by firing lasers into thin air



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<http://www.afp.com/>

3D Display / 3D Television

- 2008-Current
- Hideki Kimura
- National Museum of Emerging Science and Innovation (Miraikan), Tokyo



2009

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2014 October



<http://www.afp.com/>

3D Display / 3D Television



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<http://www.afp.com/>

3D Display / 3D Television

- Miraikan is located in Odaiba, Tokyo



www.whitehouse.gov



www.japan-guide.com

New Viewpoint Analysis

Prof. Takeo Kanade,
Carnegie Mellon University (CMU),
United States

Super Bowl

- Football fans treated to a unique new view of the action during playbacks
- CBS Television used a new technology
- Carnegie Mellon University computer vision expert, Prof. Takeo Kanade

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Super Bowl

- “Eye Vision”
- Involves shooting multiple video images of a dynamic event, such as a football game, from multiple cameras placed at different angles
- The video streams from these cameras are combined by computer and the resulting images reach viewers in a format that will make them feel as if they are flying through the scenes they see.



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Eye Vision from Prof. Takeo
Kanade, CMU

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The Matrix



www.imdb.com

- “Virtualized Reality”
- People will be able to customize the perspective from which they watch, for example, from that of a particular object.

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New Viewpoint Analysis

Augmented Reality

What is Augmented Reality (AR)?



- A combination of a real scene viewed by a user and a virtual scene generated by a computer that augments the scene with additional information.
- Generates a composite view for the user.

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What is the Goal of AR?

- To enhance a person's performance and perception of the world
- But, what is the ultimate goal?



["Augmented reality in urban places: contested content and the duplicity of code."](#) Transactions of the Institute of British Geographers, 2012

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The Ultimate Goal of AR

- Create a system such that no user CANNOT tell the difference between the real world and the virtual augmentation of it.



AR in Commerce *Geek.com*

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Augmented Reality vs. Virtual Reality

Augmented Reality

- System augments the real world scene
- User maintains a sense of presence in real world
- Needs a mechanism to combine virtual and real worlds

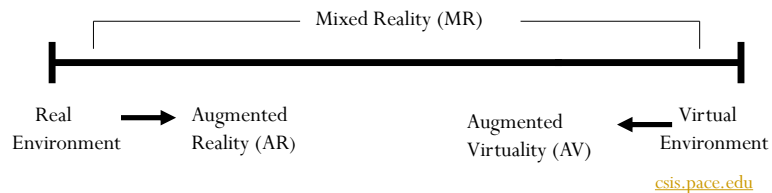
Virtual Reality:

- Totally immersive environment
- Visual senses are under control of system

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Miligram's Reality-Virtuality Continuum



Milgram coined the term “Augmented Virtuality” to identify systems which are mostly synthetic with some real world imagery added such as texture mapping video onto virtual objects.

Augmented Reality is closest to the real world because mainly a user is perceiving the real world with just a little computer generated data.

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Combining the Real and Virtual Worlds

We need:

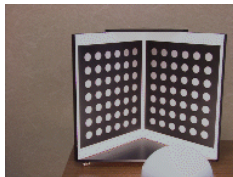
- Precise models
- Locations and optical properties of the viewer (or camera) and the display
- Calibration of all devices
- To combine all local coordinate systems centered on the devices and the objects in the scene in a global coordinate system

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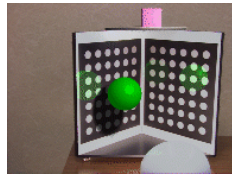
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Combining the Real and Virtual Worlds (cont)

- Register models of all 3D objects of interest with their counterparts in the scene
- Track the objects over time when the user moves and interacts with the scene



real world



real world with virtual objects and inter-reflections and virtual shading

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Mixed Reality

- Human pacman: A mobile wide-area entertainment system based on physical, social, and ubiquitous computing



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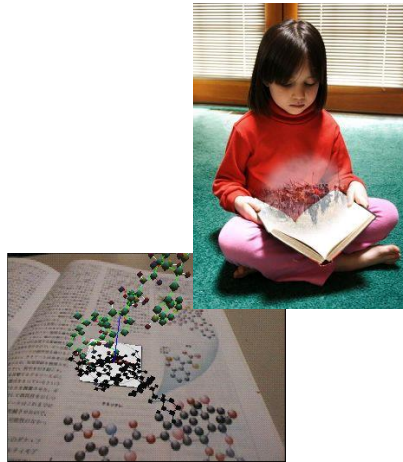
Dr. Adrian Cheok,
National University of Singapore (NUS)
at Singapore

Broadcast in Discovery Channel

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This image depicts an artist's conception of a mixed reality enhanced book. The reader is able to read the actual book, but simultaneously see three dimensional computer graphics co-located with, and in registration with the book.



www.sony CSL.co.jp

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Realistic Merging

Requires:

- Objects to behave in physically plausible manners when manipulated
- Occlusion
- Collision detection
- Shadows

****All of this requires a very detailed description of the physical scene**

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Research Activities

- Develop methods to register the two distinct sets of images and keep them registered in real-time
 - New work in this area has started to use Computer Vision (CV) techniques
- Develop new display technologies for merging the two images

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Performance Issues

Augmented Reality systems are expected:

- To run in real-time so that the user can move around freely in the environment
- Show a properly rendered augmented image

Therefore, two performance criteria are placed on the system:

- Update rate for generating the augmenting image
- Accuracy of the registration of the real and virtual image

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Failures in Registration

Failures in registration due to:

- Noise
 - Position and pose of camera with respect to the real scene
 - Fluctuations of values while the system is running
- Time delays
 - In calculating the camera position
 - In calculating the correct alignment of the graphics camera

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Applications

- Medical
- Military Training
- Hazard Detection
- Engineering Design
- Robotics and Telerobotics
- Training
- Sports
- Tourism
- Video conferencing

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Applications (cont.)

- Entertainment, Computer Games
- Manufacturing, Maintenance, and Repair
- Consumer Design
- Audio
- Education
- Architecture
- Human welfare

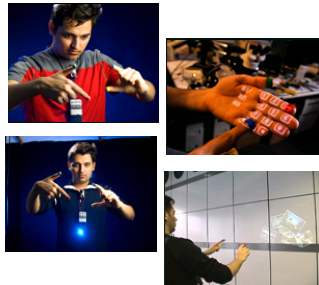
Recent Augmented Reality Applications

Sixth Sense

Sixth Sense

- A wearable gestural interface that augments the physical world around us with digital information
- Let us use natural hand gestures to interact with that information
- The SixthSense prototype is comprised of a pocket projector, a mirror and a camera.

Sixth Sense in 21st Century



Dr. Pattie Maes,
Massachusetts Institute of Technology
(MIT), United States

Broadcast in TED

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TED
TALKS

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Detail of Sixth Sense

- Sixth Sense allows people to use the internet without a screen or a keyboard
- It can turn any surface into a touch screen
- It is made with store-bought components
- *“The software program processes the video stream data captured by the camera and tracks the locations of the colored markers [that you wear on your fingers] using simple computer-vision techniques”..from MIT Media Lab*

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Detail of Sixth Sense

- The device recognizes items in stores, and then gathers and projects information about products to the user.
- It also is able to provide quick signals to let users know what the best products are based on their needs.”
- Recognize articles in newspapers, retrieve the latest related stories or video from the Internet and then display them on pages for the user
- Say goodbye to your encyclopedia

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Cost of Sixth Sense

- Only \$350 dollars, around 12,000 baht
- From an educator’s stand point, this is minimum given the amount of money spent on textbooks, computer labs and LCD projectors.
- For educator’s, Sixth Sense would be a way to put technology (literally) into each student’s hands, because of its size and its cost.

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Augmented Reality Guitar

Augmented Reality Guitar



Dr. Chutisant Kerdvibulvech,
Keio University, Japan

Broadcast in Discovery Channel



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Thank you

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National Institute of Development Administration (NIDA)
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